COURSE TITLE	**: MODERN PRODUCTION PROCESS**
COURSE CODE	**: 6111**
COURSE CATEGORY	**: A**
PERIODS PER WEEK	**: 6**
SEMESTER	**: 6**
PERIODS PER SEMESTER	**:90**
CREDITS	**: 5**

TIME SCHEDULE

MODULE	TOPIC	PERIODS
1	Understand different non-conventional Processes	22
2	Introduction to NC/CNC Machines	23
3	CNC Part Programming	22
4	Introduction of CIM ,Robotics	23
	TOTAL	90

COURSE OUTCOMES

After completion of the study the students will be able to: -

1.Understand Non conventional machining process.

2.Know NC/CNC/DNCmachines.

3.Know the program procedures.

4.Understand the principles of computer integrated manufacturing.

5.Understand Robotics.

SPECIFIC OUTCOMES

MODULE I

1.1.0 Understand different Non-conventional processes

1.1.1 Explain different Non-conventional process

1.1.2 Describe the principles of electric discharge machining

1.1.3 Explain electrical discharge grinding

1.1.4 Illustrate Electro chemical machining

1.1.5 Explain Abrasive jet machining

1.1.6 State the principle of Laser beam machining

1.1.7 Explain Wire cut Electric discharge machining

1.1.8 Describe Ultra sonic machining

<u>MODULE II</u>

2.1.0 Analyze NC/CNC/DNC Machines

2.1.1 State the principles of numerical control machine tools

2.1.2 Explain the various components with block diagram of N.C machine

2.1.3 List the steps in operation of NC machine

2.1.4 Describe the different methods of control

2.1.5 Illustrate with line sketches the point to point positioning, straight line system and contour system

2.1.6 Compare open loop system and closed loop system

2.1.7 Identify the computer numerical control (CNC) machine

2.1.8 Differentiate NC and CNC

2.1.9 Illustrate main features of CNC, DNC and CMM

2.1.10 Explain Flexible Manufacturing System (FMS)

<u>MODULE III</u>

3.1.0 Know the program procedure

3.1.1 Define G-code and M-code

3.1.2 Illustrate programming methods

3.1.3 Justify the selection of general tool information

3.1.4 Describe circular and linear interpolation

3.1.5 Explain the use of subroutines

3.1.6 Estimate programs on turning and milling

MODULE IV

4.1.0 Comprehend the Principle of Computer integrated manufacturing

4.1.1 Describe Computer integrated manufacturing system

4.1.2 Identify the features of CIM

4.1.3 Define CAD

4.1.4 Explain CAD hardware and CAD software

4.1.5 Define CAM

4.1.6 State the functions and benefits of CAM

4.1.7 Illustrate the concepts of CAD/CAM

4.2.0 Understand Robotics

4.2.1 Define robot

4.2.2 State and identify components robots

4.2.3 Classify robots

4.2.4 Illustrate working of Manipulator

4.2.5 Explain end efforts and applications

4.2.6 Identify application of robots

4.2.7 Justify the role of robots in CIMS

CONTENT DETAILS

MODULE I

1. Non-conventional machining methods

Electrical discharge machining (EDM) – introduction, machine types – contact initiated, spark initiated and electrolytic – surface finish and accuracy – EDM electrodes – advantages and limitation of EDM – EDM accessories. Electric Discharge Grinding (EDG) – introduction – application Electro Chemical Machining (ECM) – principle- surface finish – application – advantages and limitations Abrasive jet machining – introduction, equipment, accuracy and surface finish, applications, advantages and limitations. Laser Beam Machining (LBM) – introduction, equipment and machining procedure – accuracy –application advantages and limitations Wire cut Electric Discharge Machining (WCEDM) – Introduction – principle – machining procedure – applications – advantage and limitations. Ultra sonic machining.

MODULE II

1. NC/CNC/DNC Machines

Numerically Controlled machine tools – introduction – steps in operation – preparation of programmed manuscript – preparation of programmed tape , insertion of the tape in the machine console classification based on control system , point-to-point system ,straight system ,contour system. Classification based on the feedback – open loop system , closed loop system.

CNC/DNC – Introduction

Elements of CNC – work-holding devices.

Differentiate NC and CNC machines – CNC part programming (simple step turning only)

Computer aided design (CAD) – introduction, components of a CAD system.

Computer aided manufacturing (CAM) – introduction, advantages of CAD and CAM.

Application area of CAD and CAM Robots – introduction , reason for using robots , various generations of robots, power drives used – methods to stop a robot – types of arm, types of grippers , advantages and application of robots.

Flexible manufacture system (FMS) – introduction, flexibility and automation, basic components of FMS – machine tools and related equipments, material handling equipment, computer control system, human labour Types of flexible manufacturing system , flexible manufacturing unit (FMU) , flexible manufacturing cell (FMC) , flexible manufacturing transfer line (FMC or FTL)

Coordinate measuring machines, introduction, main features of CNC, CMM . Scanning, digitization, advantages

MODULE III

CNC Program procedure-coordinate system-reference point – zero point

Preparatory and miscellaneous functions – G code and M code

Method of part programming – manual and APT programming.

Tool information, speed –data, Interpolation – linear and circular, macros- subroutines , canned cycles , mirror images , thread cutting cycles, Programming practice problem on turning and milling jobs , Tool nose radius compensation – APT program

MODULE IV

Principles of CIM

Computer Integrated manufacturing – introduction, main activities under CM

CAD- Definition, benefits and stages CAD hardware – input output devices ,display devices.

Types of CAD system- CAD software – CAD features – importance Computer Communication – graphic workstation CAM – definition – functions and benefits Integrated CAD/CAM organization concept Computer integrated production system – features and advantages

Robotics

Definition of robots – classification, features, necessity.

Components of robots – illustration, its functions

Manipulator – illustration – degree of freedom

End effecters – types with illustration necessity and applications

Industrial application of robots – Advantages of robots

Artificial intelligence - introductive treatment only.

TEXT BOOKS

1 A text book of Production Engineering - P.C. Sharma

2 A text book of Production Engineering - P.C. Sharma

3 . CAD/CAM CIM – Radhakrishnan

4 . Manufacturing process for engineering materials - Serope Kalpakjian

5. Production Technology - R.K. Jain & S.C. Gupta

REFERENCE BOOKS

1. A text book of Production Technology - P.C. Sharma

2. Jigs and Fixtures - Franklin.D.Jones

3. Robotics for Engineers - Yoram Koren

4. Production Technology - HMT

5. Numerical Machine Tools - S.J Martin

6. N.C. Machines and CAM - Pressman Williams

7. Tool Design - Donaldson

8. Work Shop Technology - K.Hajra Chandhuri

9. Computer Aided Design/ Computer Aided manufacturing

- Suresh Dalela & P.K. Jain

10. CNC machines new age - VS Pabla and M. Adithyan

11. Computer integrated Design & Manufacturing - Bed Worth David, McGrawHill

12. Computer integrated manufacturing - Paul G Ranky (PHI)

13 Industrial Robotics - Golden N Nair